

All-at-Once: Modular 4D Gaussian Worldbuilding and RealityFX for Live Music Performance

Nix Liu Xin
Play.Work Labs
USA
Onceness Intelligence (OI)
USA
Harvard University
USA
nix@onceness.com

Jingyuan Wu
Independent Artist
Harvard University
USA
jingyyywu@gmail.com



Figure 1: All-at-Once: Modular 4D Gaussian Worldbuilding and RealityFX for Live Music Performance

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Real-Time Live! '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2135-9/2025/12

<https://doi.org/10.1145/3757375.3774346>

ACM Reference Format:

Nix Liu Xin and Jingyuan Wu. 2025. All-at-Once: Modular 4D Gaussian Worldbuilding and RealityFX for Live Music Performance. In *SIGGRAPH Asia 2025 Real-Time Live! (SA Real-Time Live! '25)*, December 15–18, 2025. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757375.3774346>

1 Overview

All-at-Once transforms the emerging field of volumography—particularly Gaussian Splatting—into a real-time, modular worldbuilding instrument for live audiovisual performance. Instead of rendering static

volumetric captures, we treat volumography as a performative medium that can be remixed, manipulated, and composed in real time.

Built upon a custom 4D Gaussian pipeline and interactive RealityFX system, the project enables performers to spatially “play” volumetric worlds as though they were musical instruments. Each musical gesture—beat, note, modulation— triggers a corresponding volumetric transformation: objects rotate, multiply, dissolve, and re-emerge in synchrony with sound. Music becomes an interface for world-making.

The result is a synesthetic experience where remixing music equals remixing reality. A simple bassline may ripple through volumetric space, while a chord shift transforms a photoreal room into

a surreal dreamscape. By bridging Gaussian Splatting, procedural modulation, and spatial music mapping, All-at-Once proposes a new grammar for live volumetric performance—one that is poetic, participatory, and technically groundbreaking.

For the SIGGRAPH audience, this demonstration illustrates a conceptual and technical leap: from reconstructing the world to performing it. It showcases the first instance of RealityFX, a real-time volumetric effects layer for Gaussian media, and establishes a new paradigm where AI, graphics, and music converge into living, generative stages of performance.

In essence, All-at-Once invites us to imagine the next generation of concerts, theater, and XR experiences—where sound and vision truly merge into one expressive, real-time volumetric art form.